

How to introduce AI into enterprise decision-making *without giving up control of the truth.*

FROM ARCHITECTURE NOTE N° 001

*“An LLM should never own your numbers.
It should explain them.”*

BA

BENACTA



BUILT ON TRUTH. DESIGNED FOR DECISIONS.

Your company already has the data. The bottleneck is the path from data to decision.

An enterprise of any scale already owns an ERP, an EPM, a CRM, a BI stack, document stores and a data platform. The data exists. The reports exist. Yet between those systems and an actual decision, people still collect, reconcile, search, interpret, explain, email, approve and follow up, every month, in every function.

None of these systems is useless. The problem is that truth lives in one place, context in another, judgment in meetings and inboxes, and action in whatever tool the owner opens next. Truth, context, judgment and action are fragmented, and the fragmentation, not the data, is the bottleneck.

WHERE MOST STOP

REPORTING
ANALYTICS

DECISION INTELLIGENCE
ACTION

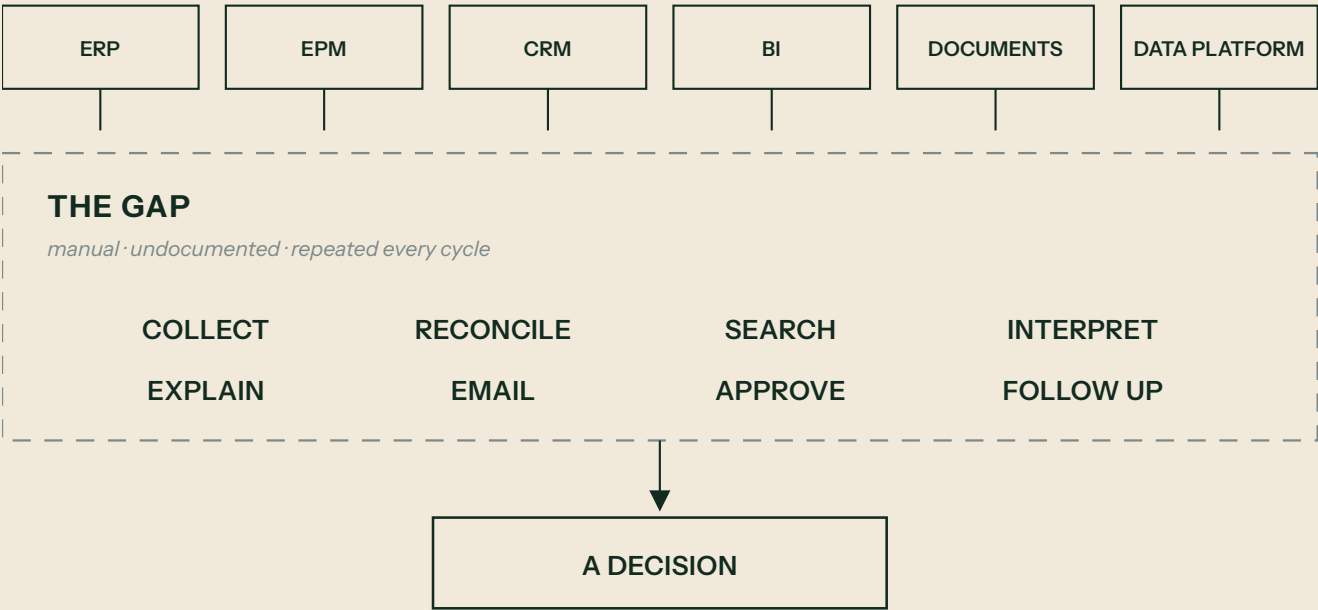
most organizations stop above the line

58%

of finance functions already use AI, up 21 points in one year.
Gartner, CFO & Finance AI Survey, Sept. 2024

adoption is no longer the question; control is

SYSTEMS OF RECORD · EACH COMPLETE, EACH SEPARATE



The opportunity is not another dashboard.
And it is not another chatbot.

From enterprise data to governed decisions.

One system, seven responsibilities, one auditable path.

A Decision Intelligence System does not replace the systems you own. It engineers what is missing between them: the path itself. Each layer has a single responsibility, each hand-off is explicit, and every step is logged.

read-only, nothing re-keyed

*the system knows what
things mean, not just totals*

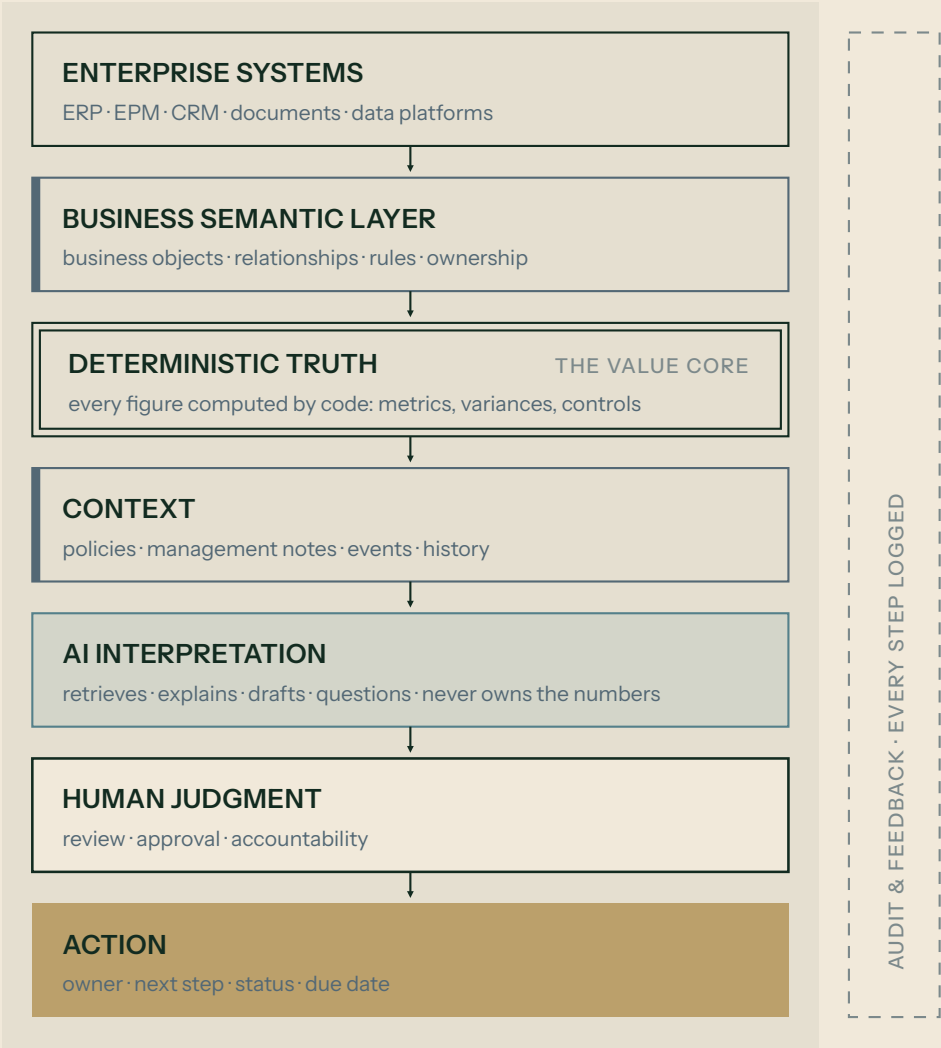
*every figure computed,
none invented*

*the note that explains
the movement, found*

*commentary in hours,
not days*

approved before it ships

*insight becomes an
owned next step*



Engineer the truth. *Augment the judgment.*

Code computes. AI explains. Humans decide.

CODE

Facts
Calculations
Metrics
Controls
Rules

AI

Retrieval
Context
Interpretation
Drafting
Questions

HUMANS

Judgment
Approval
Accountability
Action

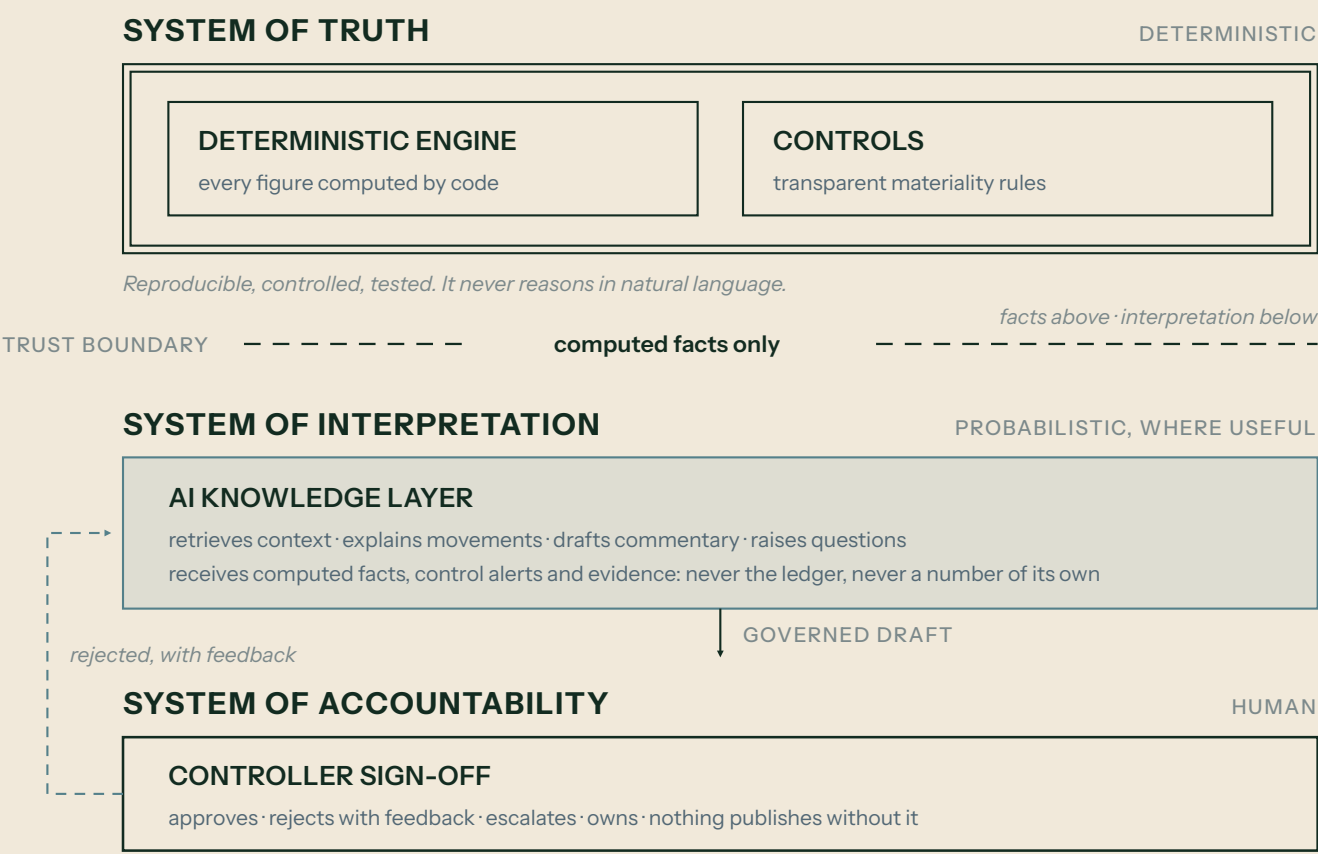
The placement is what makes AI usable in an enterprise: probabilistic interpretation where it adds value, deterministic code where the figures must be reproducible, human judgment where accountability lives. Each layer does the work it is built for, and only that work.

DATA → BUSINESS OBJECTS → TRUTH → CONTEXT → INTERPRETATION → DECISION → ACTION → OUTCOME



Not every part of the system should be probabilistic.

A decision system contains three kinds of components, and the architecture must keep them apart. The trust boundary is a working design rule with one contract: **only computed facts cross the boundary.**



“The LLM is not the system. It is one component inside the system.”

THE REMOVABILITY TEST

Switch the AI off, and every figure, every control and every materiality classification must be unchanged, because none of it was ever the AI’s to compute. In the accompanying reference implementation this is a permanent automated test, not a claim.

Technical note: `test_financial_truth_is_independent_from_llm`. The truth layer has no import path to any AI component, and the full pipeline is re-run with those paths blocked.

Decisions happen around business objects, not database rows.

A ledger row records that an amount moved. It does not know that the amount belongs to a project; that the project serves a customer under a contract; that revenue is recognized when milestones are accepted; or that a named person owns the outcome.



Every line is a relationship a decision depends on: who the work is for, what governs it, who owns it.

A ROW

2026-07 · 700100 · BU-PRJ · CC-4100 · 910,000

records that an amount moved

A BUSINESS OBJECT

what it represents · what it relates to
which rules apply · who owns it · what can be decided
carries the meaning a decision needs

A **Business Semantic Layer** makes that knowledge part of the system: what an object represents, what it relates to, which rules apply to it, who owns it, and which decisions can be taken around it. Only then is AI given something worth interpreting, and only then can an insight land on the person who can act on it.

In July, two customer acceptance milestones slipped, on two different projects, for two different customers. Because each milestone is connected to its project, its revenue and its owner, the shortfall resolves to named work with a named owner rather than to a number. The next page shows that month.

Technical readers know this layer as a lightweight ontology or domain model. The reference implementation keeps it small (objects, relationships, metric definitions, ownership), sized to one decision process, and it rejects source rows that disagree with the governed model rather than silently aggregating them.

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One material variance, end to end.

REVENUE · JULY FY26

ACTUAL	€4.72M
BUDGET	€5.00M
VARIANCE	-€280K
VARIANCE %	-5.6%

HIGH · MATERIAL

Every figure on this page is computed by code from the general ledger and the FY26 budget: never estimated, never generated.

WHAT HAPPENED?

Revenue finished €280,000 below budget for the month.

WHY?

Two customer acceptance milestones planned for July were rescheduled into August at the customer's request. Neither reschedule reflects a scope change, a commercial dispute or a quality issue, and both contracts remain in force.

WHAT IS THE EVIDENCE?

- ✓ FY26 Forecast Assumptions · *Revenue Recognition Basis*
- ✓ Management Commentary July 2026 · *Projects Business Unit*
- ✓ Project Milestone Register FY26 · *July 2026 Milestone Status*

Retrieved by the system, with the reason each passage was selected, not a curated list.

WHAT REQUIRES ATTENTION?

The variance exceeds the company's €100,000 absolute materiality threshold. That threshold is written policy the system can cite, not an opinion.

WHAT COULD WE DO NEXT?

Review milestone recognition with Project Finance. *A suggested follow-up for a human, never labeled a decision.*

WHO APPROVES?

AI-generated interpretation · **Controller approval required.** Until a named controller approves it, the interpretation is a draft and nothing downstream may use it. Requesting a revision is a designed path, not an error.

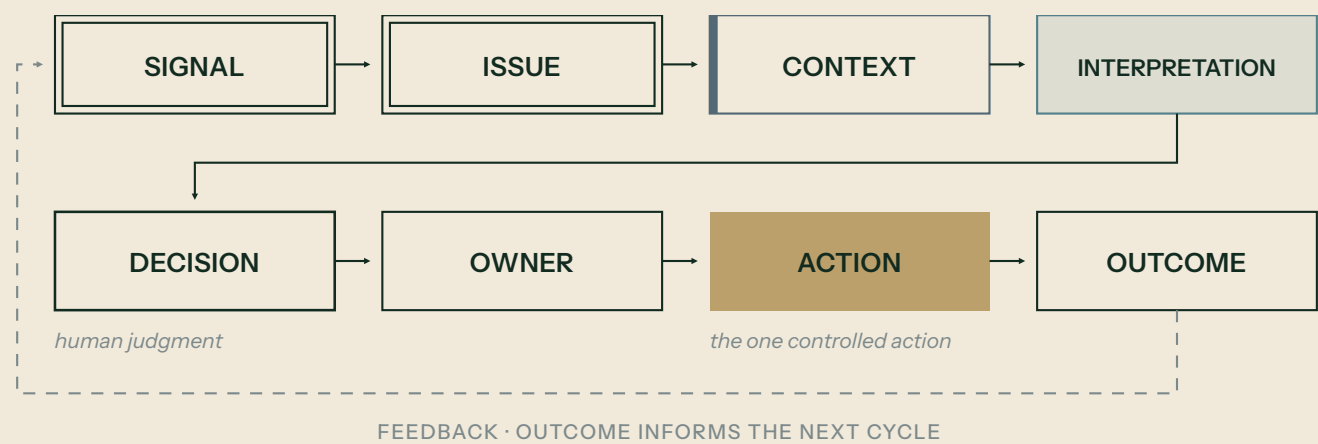
WHO OWNS THE NEXT STEP?

Project Finance. The owner is proposed by the system alongside the finding and assigned by the reviewer, never assigned automatically.

All data is fictional: Meridian Industrial Group is a reference scenario built for this Blueprint and its accompanying implementation. The figures above are produced by the running system and verified by its automated tests.

A dashboard ends at insight. A decision system continues into action.

Insight without action is incomplete. In a decision system, an accepted interpretation does not stop at “noted”; it becomes an issue with an owner, a next step, a status and a due date, and the loop closes when the outcome is recorded.



FROM THE DECISION LOG: ONE ISSUE, CARRIED ALL THE WAY THROUGH

ISSUE	Travel above budget: €38,000 over plan (+31.1%), unplanned customer workshops
REVIEWED BY	A. Controller · “Consistent with the travel policy; workshops were customer-driven.”
OWNER	Cost Center Manager · Projects
NEXT STEP	Confirm workshop travel against policy and open the missing budget line.
STATUS	ACTION REQUIRED · due 21 August 2026

THE REFERENCE IMPLEMENTATION

demonstrates this loop at its smallest honest size:
issue → reviewer → owner → next step → status,
every step logged. It is a working design pattern, not
a workflow platform.

THE ENTERPRISE DIRECTION

connects the same loop to the task, approval and
workflow systems a company already runs, and
measures outcomes (decision latency, cycle time,
adoption) over time.

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Start with one decision.

Prove it on one cycle, six to eight weeks, then scale.

EXAMPLE BASELINES

Top-quartile finance functions
close in **5 days or fewer**
APQC, Open Standards
Benchmarking

and run at **0.55% of revenue**
against a ~1% median
PwC, Finance Effectiveness
Benchmarking 2024

*your first measurement is your own
current cycle*

01

PICK ONE DECISION PROCESS

Recurring, material, with a named owner. The monthly performance review is a natural first candidate.

02

MAP THE BUSINESS OBJECTS

The entities and relationships the decision is actually about, and who owns each of them.

03

ENGINEER THE TRUTH

Data, metrics, rules and controls, computed by code and reconciled to source. This is the value core.

04

ADD CONTEXT

The policies, notes, events and history that explain why a movement happened.

05

ADD AI INTERPRETATION

Explanation, comparison, drafting, open questions. Computed facts in; a governed draft out.

06

KEEP HUMANS IN CONTROL

Review, approval and escalation as designed paths; rejection is a feature, not a failure.

07

CONNECT TO ACTION

Every accepted insight gets an owner, a next step and a status you can follow up on.

08

MEASURE OUTCOMES

Against your own baseline: cycle time, accuracy, decision latency, analyst capacity, adoption, business result.

Truth before interpretation. Humans before action. Steps 03, 05 and 06 are the order that makes the rest safe.

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Don't start with AI. *Start with the decision.*

PROOF, NOT PROMISES

A working reference implementation accompanies this Blueprint and demonstrates every layer described here: the Business Semantic Layer, the Deterministic Core, the Control Engine, Context Retrieval, AI Interpretation, Human Review, Decision & Action, and the Audit Trail. It runs entirely without an AI key because the truth never depended on one.

EXPLORE THE REFERENCE IMPLEMENTATION

github.com/anasbenazzouz/benacta-decision-intelligence-blueprint

FOR DECISION-MAKERS

If you are choosing the first decision process to prove this on, BENACTA can help you scope it.

Website · LinkedIn: *to be added*

FOR YOUR ARCHITECTS

Send them the reference implementation. The boundaries described in this Blueprint are enforced there by tests, not by slideware.

Live demo: benacta-di.streamlit.app

BENACTA

Enterprise AI · Decision Intelligence

AI-Native Decision Systems

Finance · Operations · Performance · Workflows

Turning enterprise data into decisions and action.

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